

MCP Configuration Reference

What is MCP?

Model Context Protocol (MCP) is an open standard for connecting AI tools to external systems. Think of it as “USB for AI”—a standardized way to plug different data sources and tools into any MCP-compatible AI assistant.

Architecture Overview

Your AI Tool (Client)	for example, VS Code + Copilot/Cline, Claude Code, Cursor
MCP Protocol	
MCP Server (local or remote)	Lightweight adapter
API/SDK	
External Service	GitHub, PostgreSQL, Figma, etc.

Configuration Locations

Claude Code (global): ~/.claude/settings.json

Claude Code (project): .claude/settings.json

VS Code + Cline: Configure in Cline settings panel

Cursor (project): .cursor/mcp.json

VS Code: .vscode/mcp.json or settings.json

GitHub MCP Server

Provides: Issues, PRs, commits, CI status, repository information

```
{
  "mcpServers": {
    "github": {
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-github"],
      "env": {
        "GITHUB_PERSONAL_ACCESS_TOKEN": "ghp_your_token_here"
      }
    }
  }
}
```

Token setup: GitHub Settings > Developer Settings > Personal Access Tokens > Generate New Token. Grant “repo” scope at minimum.

PostgreSQL MCP Server

Provides: Schema queries, table inspection, SQL execution

```
{
  "mcpServers": {
    "postgres": {
      "command": "npx",
      "args": ["-y", "@modelcontextprotocol/server-postgres"],
      "env": {
        "POSTGRES_CONNECTION_STRING": "postgresql://user:pass@localhost:5432/mydb"
      }
    }
  }
}
```

Custom MCP Server (TypeScript)

Create your own MCP server for internal APIs:

```
import { McpServer } from "@modelcontextprotocol/sdk/server/mcp.js";
import { StdioServerTransport } from
"@modelcontextprotocol/sdk/server/stdio.js";

const server = new McpServer({
  name: "my-custom-server",
  version: "1.0.0",
});

// Define a tool
server.tool("tool-name", "Description of what this tool does", {
  paramName: { type: "string", description: "Parameter description" }
}, async ({ paramName }) => {
  // Fetch data from your API
  const response = await fetch(`https://your-
api.internal/endpoint/${paramName}`);
  const data = await response.json();
  return {
    content: [{ type: "text", text: JSON.stringify(data, null, 2) }]
  };
});

// Start
const transport = new StdioServerTransport();
await server.connect(transport);
```

Register the custom server:

```
{
  "mcpServers": {
    "my-server": {
      "command": "npx",
      "args": ["tsx", "./mcp-servers/my-server.ts"]
    }
  }
}
```

Available MCP Servers (Popular)

GitHub (@modelcontextprotocol/server-github)—Issues, PRs, commits, CI status

PostgreSQL (@modelcontextprotocol/server-postgres)—Schema, queries, data inspection

MySQL (@modelcontextprotocol/server-mysql)—Schema, queries, data inspection

Filesystem (@modelcontextprotocol/server-filesystem) — Extended file operations

Puppeteer (@modelcontextprotocol/server-puppeteer)—Browser automation, screenshots

Sentry (Community server)—Error tracking, issue investigation

Slack (Community server)—Team communication, channel queries

Linear/Jira (Community servers)—Issue tracking, sprint management

Security Best Practices

- Use read-only database credentials for MCP connections (SELECT-only permissions).
- Store credentials in environment variables, not in config files committed to git.
- Add MCP config files to .gitignore if they contain credentials.
- Never connect MCP to production databases without careful access restrictions.
- Start with read-only MCP servers; add write operations only with confirmation steps.
- Limit token scopes to the minimum required (e.g., 'repo' only for GitHub).

Troubleshooting

MCP server not connecting: Restart your AI tool after adding/modifying MCP config. Check that npx can find the package.

Authentication errors: Verify your token/credentials are correct and have sufficient permissions.

Tools not appearing: Some tools may take a moment to load. Check the AI tool's MCP status indicator.

Performance issues: Too many MCP servers can dilute context. Start with one or two and add more as needed.